

SWAPNA

AI ENGINEER / DEEP LEARNING SPECIALIST (UNDERGRADUATE)

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PROFESSIONAL SUMMARY

Aspiring Artificial Intelligence Engineer currently in the 4th year of a Bachelor of Technology program. Hands-on experience architecting and building deep learning pipelines for computer vision and smart city infrastructure. Adept at transforming raw sensor and video feeds into actionable edge intelligence, specifically demonstrating engineering capabilities in object detection (YOLO), sequential analysis (Optical Flow), and full-stack software prototyping for social impact solutions. Proven hackathon performer skilled at taking AI concepts from ideation to functional prototypes.

TECHNICAL SKILLS

AI & Deep Learning:	Computer Vision, Object Detection (YOLOv8/v10), Optical Flow, CNNs, Sequence Modeling
Languages:	Python (Advanced), SQL, C++, HTML5, CSS3
Frameworks & Libs:	PyTorch, OpenCV, NumPy, Pandas, Scikit-Learn, Flask, FastAPI
Tools & Cloud:	Git/GitHub, Docker, Google Colab, VS Code, Weights & Biases
Key Concepts:	Real-time Video Processing, Pipeline Optimization, Data Annotation, Dashboard Mockups

TECHNICAL PROJECTS

AI-Powered Real-Time Automatic Road Accident Detection System

Dec 2025 – Jan 2026

Core Deep Learning & Pipeline Architect

Academic Project

- Designed a multi-stage intelligent computer vision pipeline integrating **YOLO object detection** with **Optical Flow localization** algorithms to process video feeds from traffic monitoring cameras.
- Engineered real-time collision detection logic to capture vehicular interactions, classify accident severity dynamically, and trigger instantaneous automated alert dispatches.
- Developed high-fidelity custom visual mockups representing live authority-facing CCTV dashboards and alert systems to illustrate complex multi-camera collision detection logic effectively.
- Optimized frame-rate throughput across the pipeline using OpenCV video streaming optimization techniques, ensuring reliable real-time edge processing capabilities.

Smart Waste Management & Sanitation Monitoring System

Jan 2026

Lead Developer (Hackathon Concept & Prototype)

Technical Hackathon

- Conceptualized and prototyped a comprehensive AI-driven civic platform empowering citizens to capture and report localized sanitation issues seamlessly via image uploads.
- Implemented an image classification and object localization back-end utilizing deep learning models to identify waste categories, estimate bin overflow thresholds, and filter noise inputs.
- Architected a central administrative authorities' dashboard interface with responsive UI components to display prioritized geo-tagged issues, dispatch queues, and analytics.
- Synthesized user workflow diagrams, operational flowcharts, and technical presentation collateral to present the product architecture convincingly to a hackathon judging panel.

ACHIEVEMENTS & LEADERSHIP

Hackathon Participant & Tech Presenter

2025 – 2026

- Collaborated in fast-paced multidisciplinary teams to engineer scalable software solutions aiming at smart city infrastructure optimization and public safety enhancements.
- Leveraged advanced prompt-engineering and visual prototyping tools to rapidly synthesize technical architectures, pipeline diagrams, and user interface mocks for investor pitching sessions.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering

Expected Graduation: June 2026

Specialization / Core Electives: Artificial Intelligence & Deep Learning

Andhra Pradesh, India

- Current Academic Standing: **4th Year undergraduate student.**
- Relevant Coursework: Deep Learning Foundations, Design and Analysis of Algorithms, Computer Vision, Database Management Systems, Advanced Python Architectures.